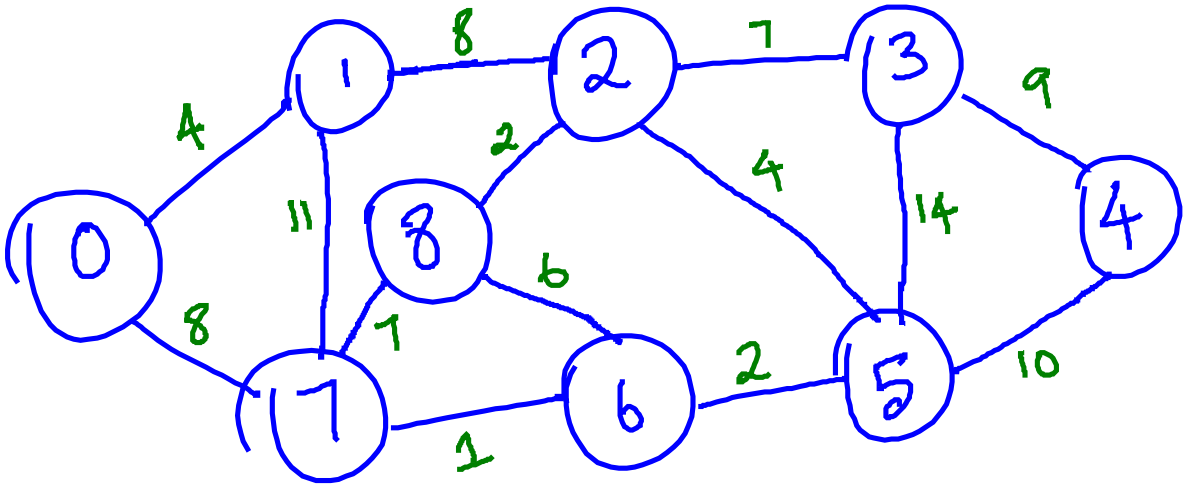
Aim:

Single Source Shortest Path



Find Shortest paths from 0 to all other nodes

→ Same Time complexity for finding Shortest path from src to dst.

→ Dijkstra's algorithm

[Edsger V. Dijkstra, Dutch scientist]

Algorithm: (Assuming graph is Connected)

known = ϕ [All vertices are Unknown]

dist (src) = 0

dist (v) = ∞, v ≠ src

dist:

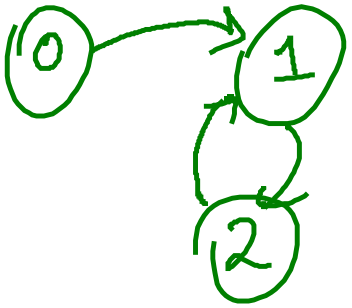
0	1	2	...	8
0	∞	∞	...	∞

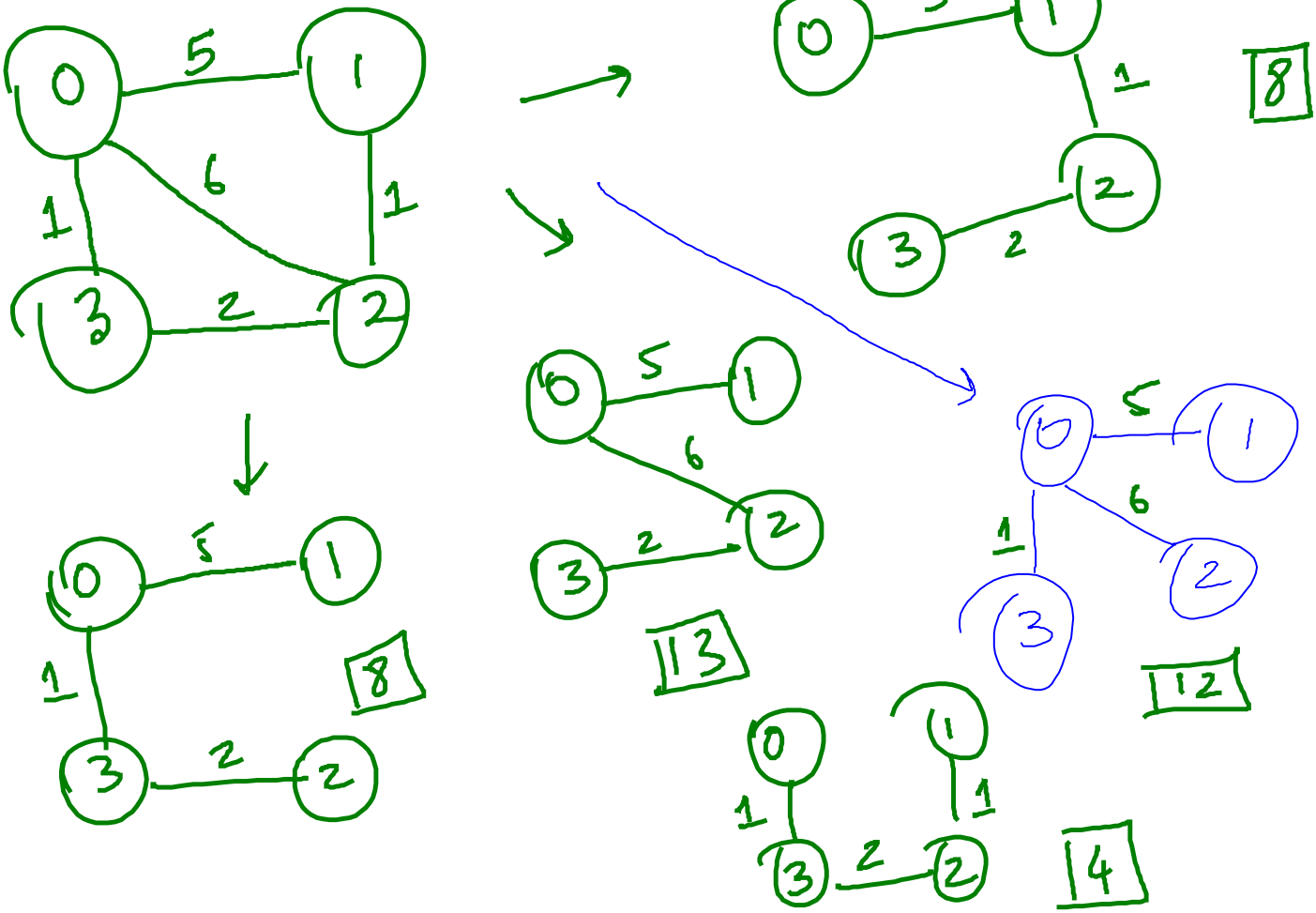
While (there are Unknown Verti

Sai

Minimum Spanning Tree :

- Tree (no cycle)
- Spanning (all vertices)
- Minimum overall cost





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Minimum Spanning T

	Known	dist
	0	
	0	∞
7	0	∞
8	10	∞

vtex	Known	dis
0	0	0
1	0	∞
2	0	∞
3	0	∞
4	0	∞
5	0	∞
6	0	∞
7	0	∞
8	0	∞

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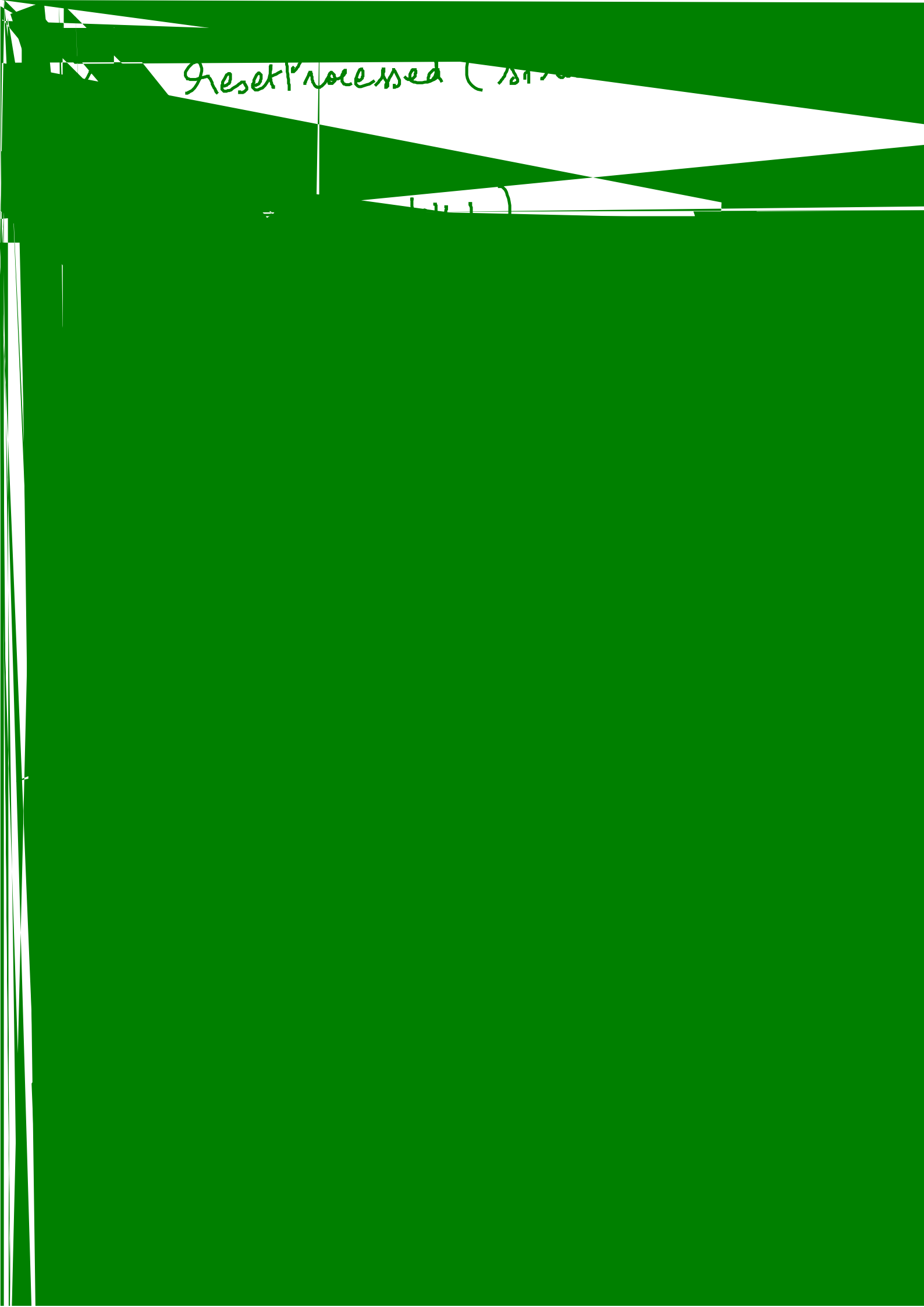
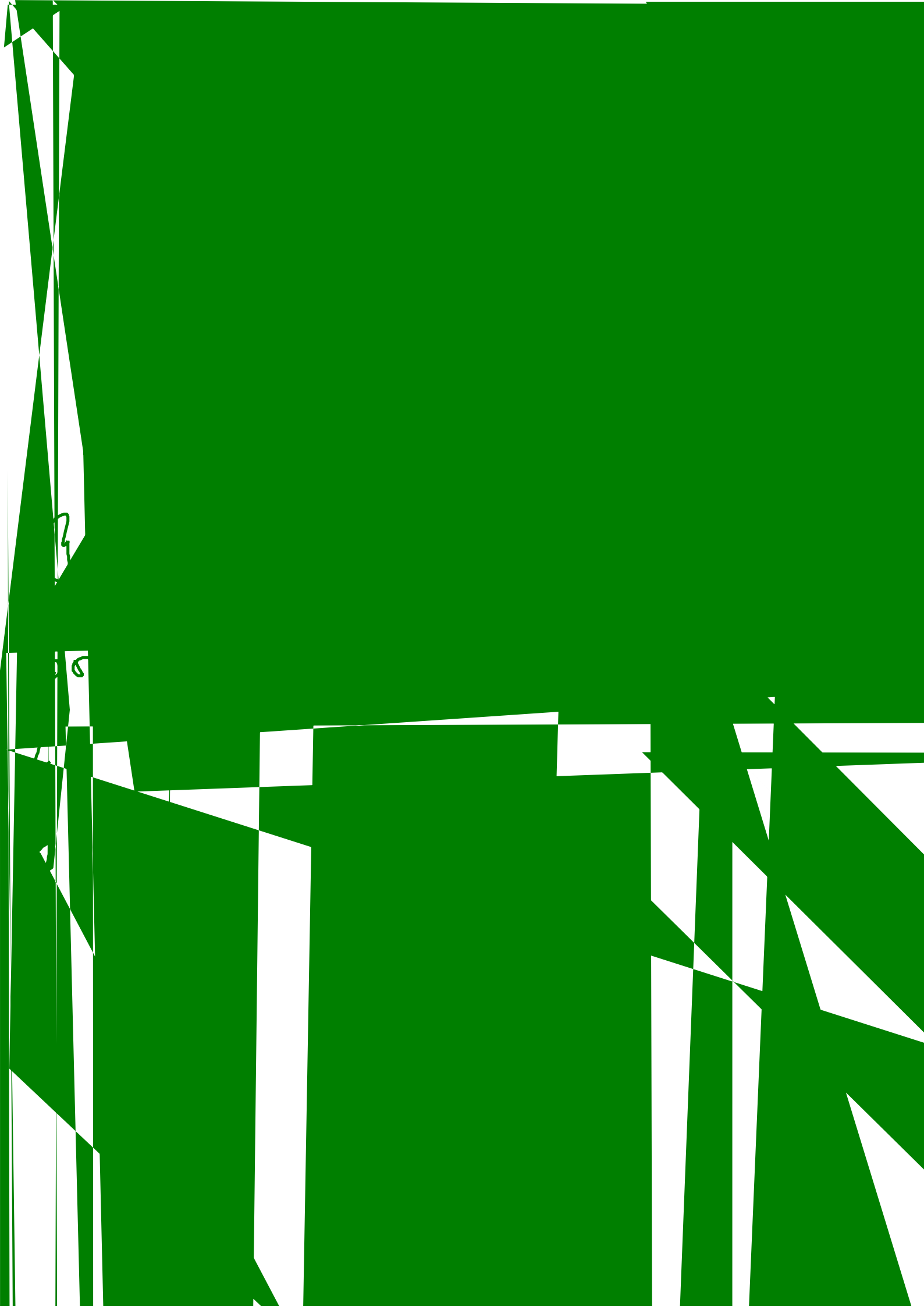
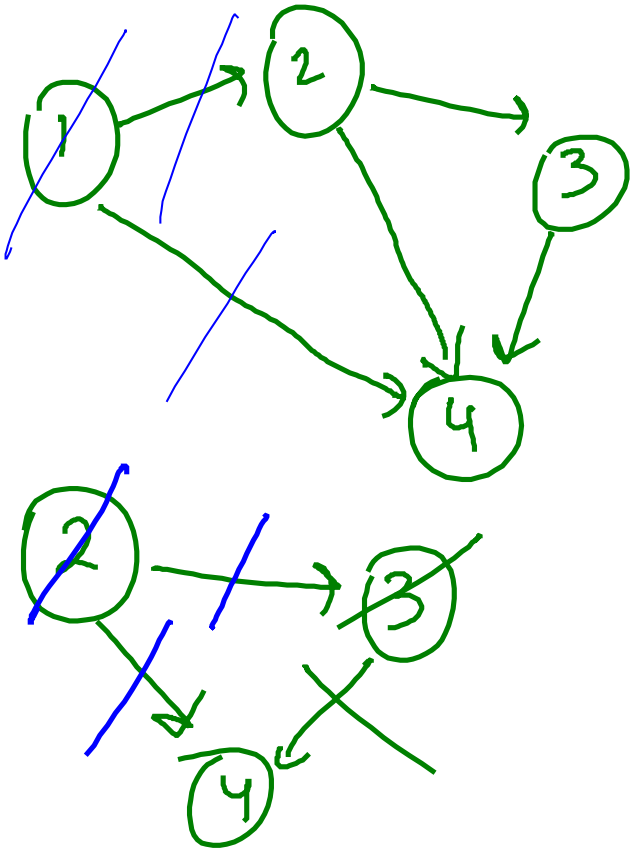
To bm

```
        a = a -> next;
    }
    } // end of while q not Empty
    resetProcessed()
} // end of function
```


ResetProcessed

bool p





1 , 2 , 3 , 4